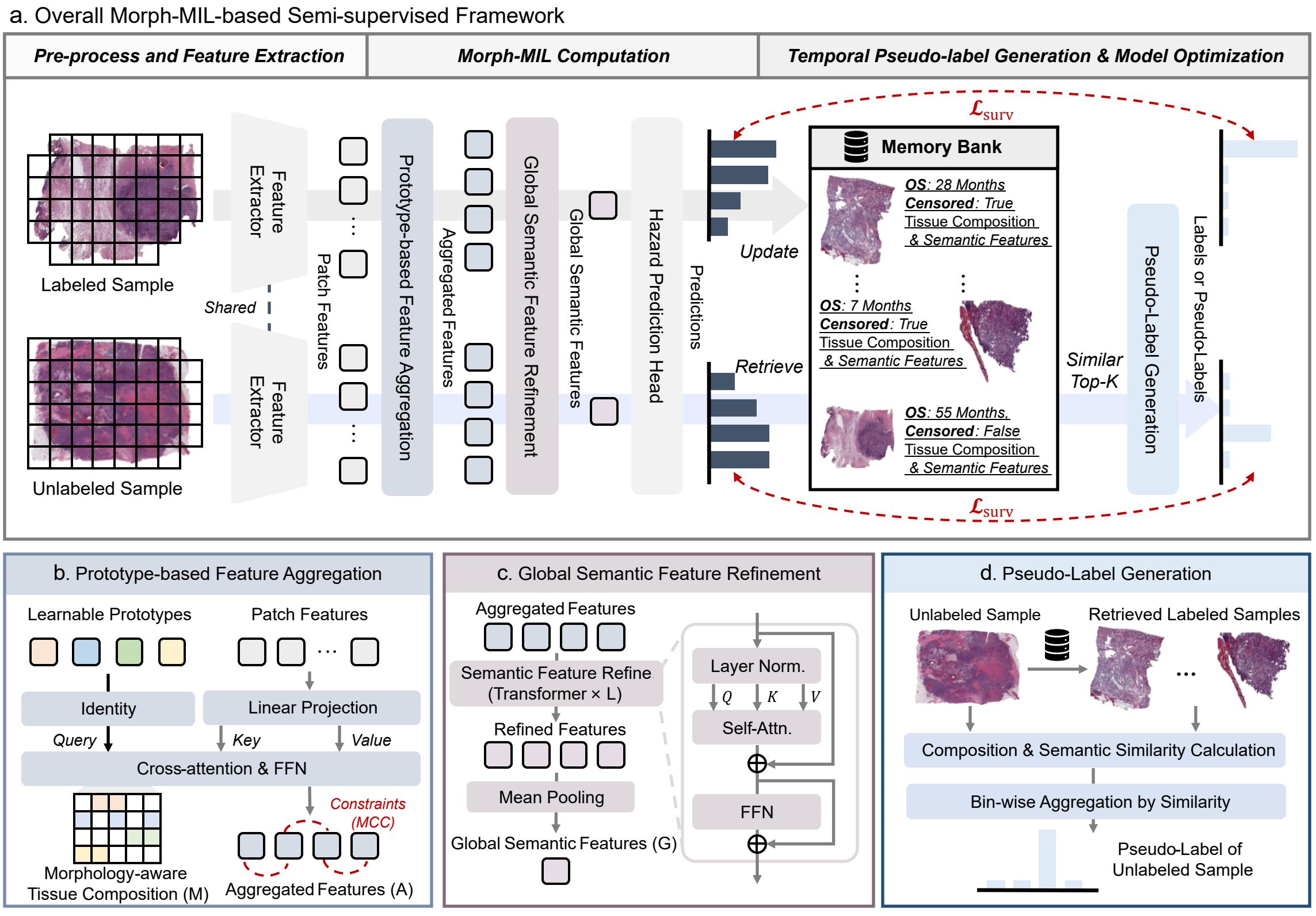
a. Overall Morph-MIL-based Semi-supervised Framework
Pre-process and Feature Extraction
Morph-MIL Computation
Temporal Pseudo-label Generation & Model Optimization
Labeled Sample
Unlabeled Sample
Shared
Feature Extractor
Feature Extractor
Patch Features
Prototype-based Feature Aggregation
Aggregated Features
Global Semantic Feature Refinement
Global Semantic Features
Hazard Prediction Head
Predictions
Update
Retrieve
Memory Bank
OS: 28 Months
Censored: True
Tissue Composition & Semantic Features
OS: 7 Months
Censored: True
Tissue Composition & Semantic Features
OS: 55 Months,
Censored: False
Tissue Composition & Semantic Features
Similar Top-K
Pseudo-Label Generation
Labels or Pseudo-Labels
$\mathcal{L}_{surv}$
$\mathcal{L}_{surv}$
b. Prototype-based Feature Aggregation
Learnable Prototypes
Patch Features
Identity
Linear Projection
Query
Key
Value
Cross-attention & FFN
Constraints (MCC)
Morphology-aware Tissue Composition (M)
Aggregated Features (A)
c. Global Semantic Feature Refinement
Aggregated Features
Semantic Feature Refine (Transformer × L)
Refined Features
Mean Pooling
Global Semantic Features (G)
Layer Norm.
Q
K
V
Self-Attn.
FFN
d. Pseudo-Label Generation
Unlabeled Sample
Retrieved Labeled Samples
Composition & Semantic Similarity Calculation
Bin-wise Aggregation by Similarity
Pseudo-Label of Unlabeled Sample